STAT 214 Spring 2026
Week 3

Zach Rewolinski

Required prep - setting up computing resources

We will use computing resources provided by NSF ACCESS, a program which gives access to several high-performance computing platforms.

In the next discussion session, I will help you access Bridges-2, a supercomputer at the Pittsburgh Supercomputing Center (PSC).

Before that happens, we need to set up your accounts.

Required prep - setting up computing resources

1. Make account with NSF ACCESS here:

    `operations.access-ci.org/identity/new-user`

2. Click “**Register with existing identity**”
3. Under “**Select an Identity Provider**”, select “**University of California, Berkeley**”
4. Click “**LOG ON**” and follow the steps to make an account.
5. Once you have an account, send email to me with your ACCESS username.
    - Email: zachrewolinski@berkeley.edu
    - Subject Line: [STAT 214] ACCESS Username
    - Body: Include your full name and ACCESS username.
6. After we add you, the PSC will send you an email with account info.

Required prep - setting up computing resources

If you somehow didn’t complete these steps, check the “Making an account” section of:

`stat-214-gsi/computing/psc-instructions.md`

🚨 Announcements 🚨

- A README has been added to the `lab1` folder of the `stat-214-gsi` repo. It summarizes what I said in discussion last week regarding the lab.
- For those with questions about `git`, please see `git-supplementary-materials.pdf` in the `discussion/week3` folder of the repo. This also includes a bit of information on hidden files such as `.gitignore`.

Any Lab 1 questions?

Overview: Singular Value Decomposition

(on whiteboard)

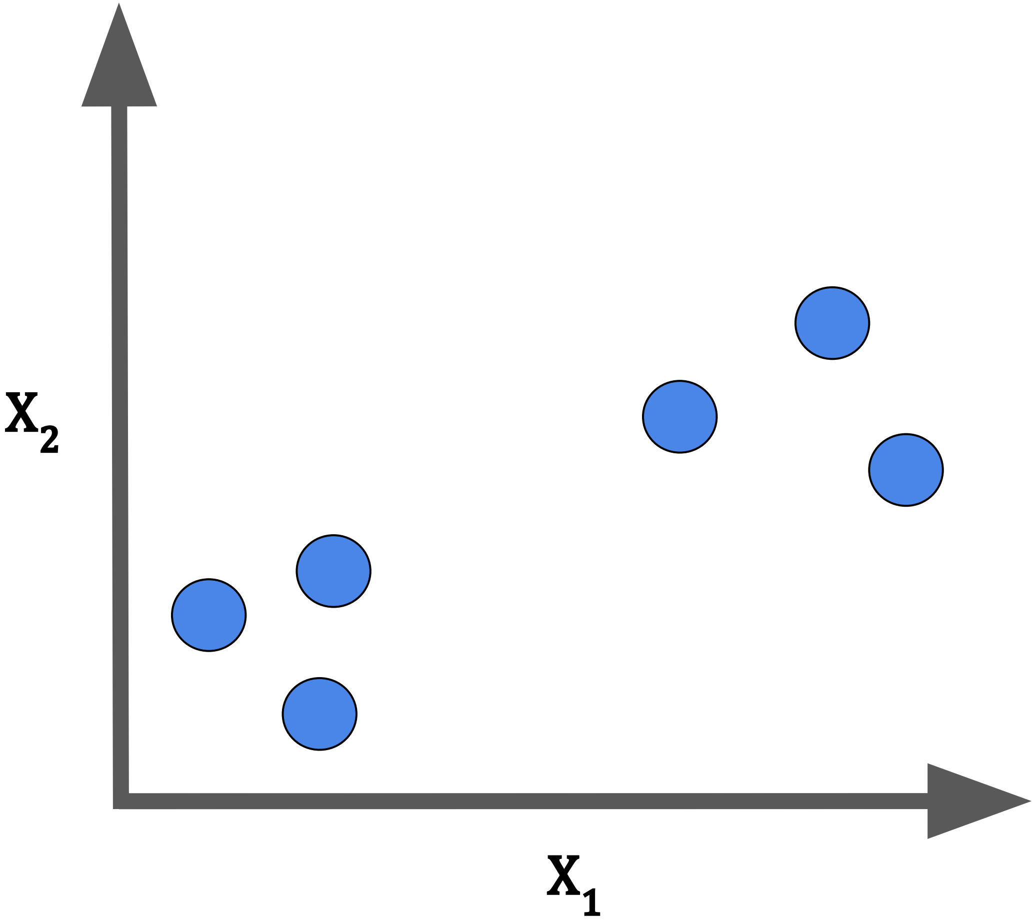
Visual Explanation: Principal Component Analysis

(on whiteboard)

Another Visual Explanation: PCA

We will once again use a simple dataset with only two features: $\mathbf{X_1}$ and $\mathbf{X_2}$.

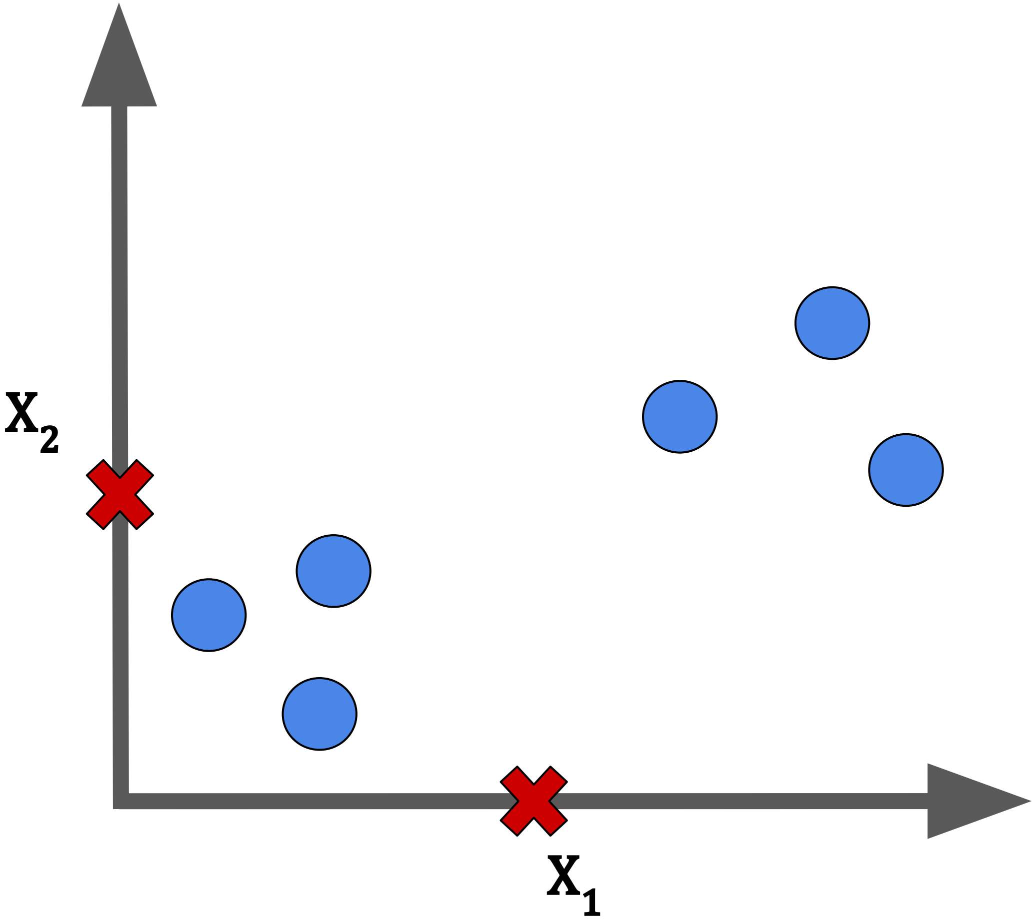
We start by plotting the data.



We will once again use a simple dataset with only two features: X_1 and X_2.

We start by plotting the data.

We then calculate the average of both features.

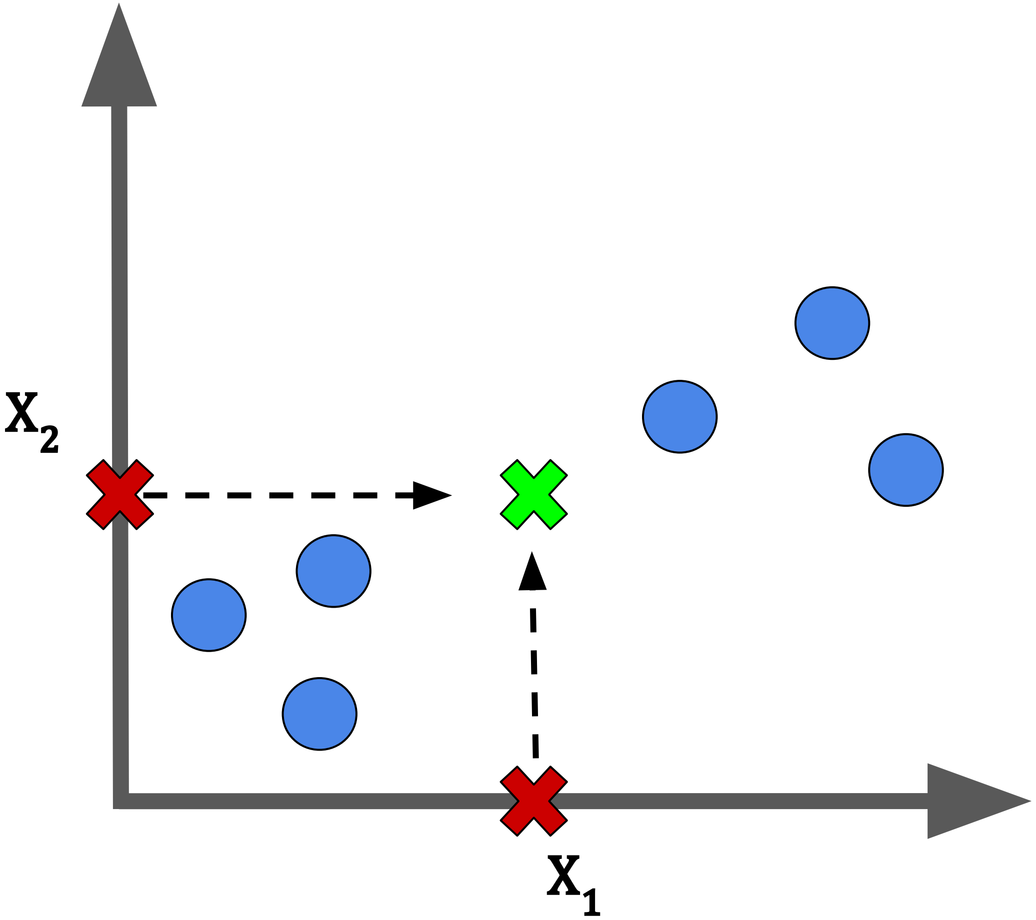


We will once again use a simple dataset with only two features: X_1 and X_2.

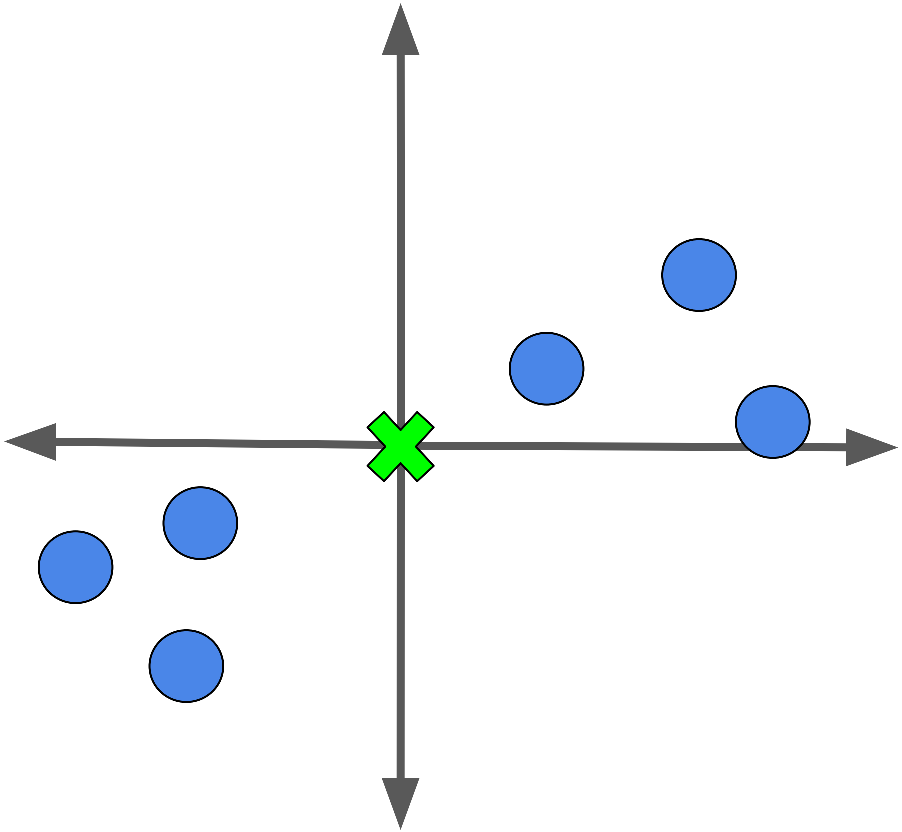
We start by plotting the data.

We then calculate the average of both features.

We can use these values to find the center of our data.

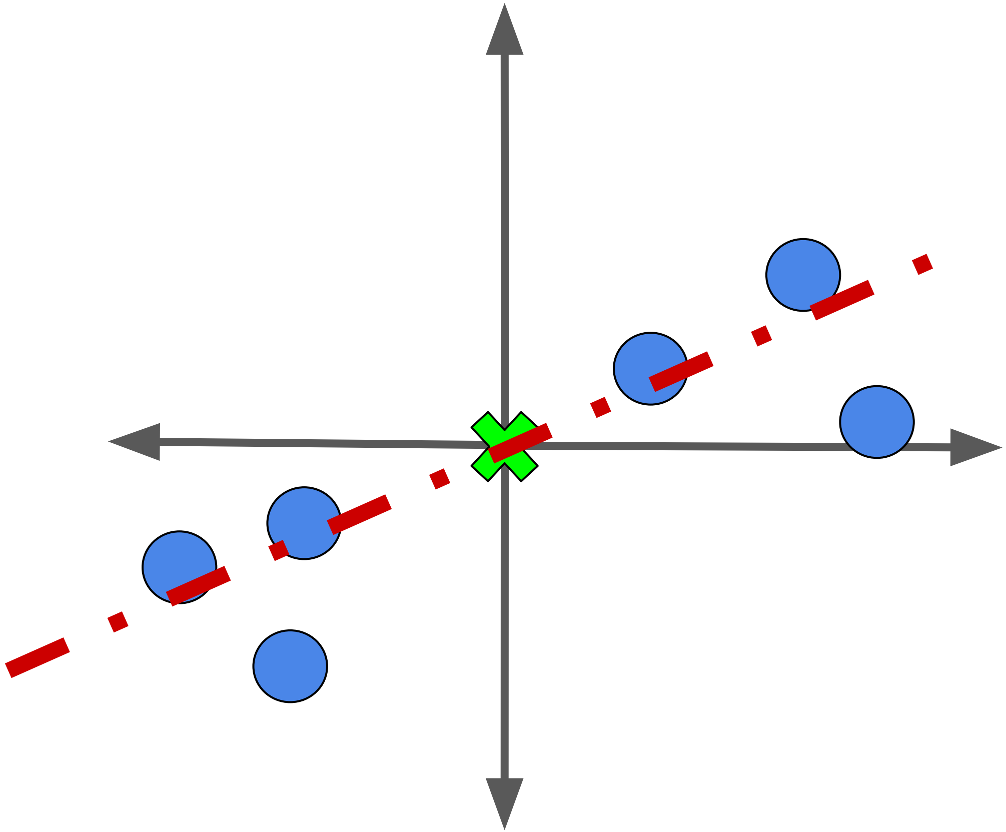


Let us shift our data so that it is centered at the origin.



Let us shift our data so that it is centered at the origin.

Now let us try something different: fitting a line to our data that goes through the origin.



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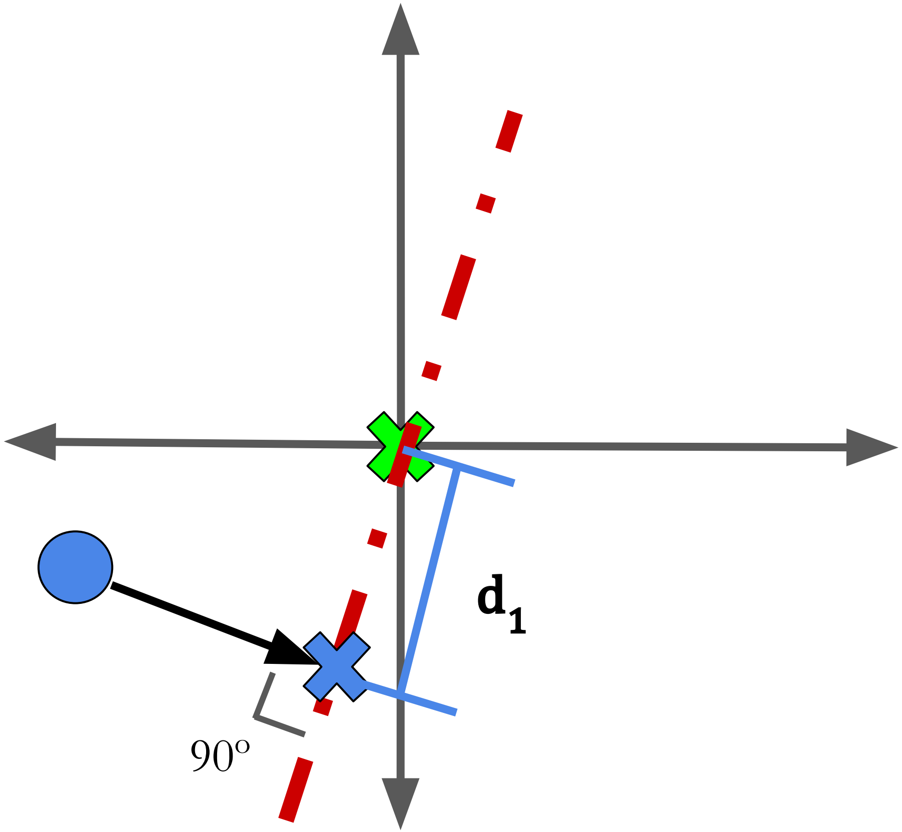
Now let us try something different: fitting a line to our data that goes through the origin.

To see that it maximizes the distance between the points and the origin, let us focus on a particular data point.



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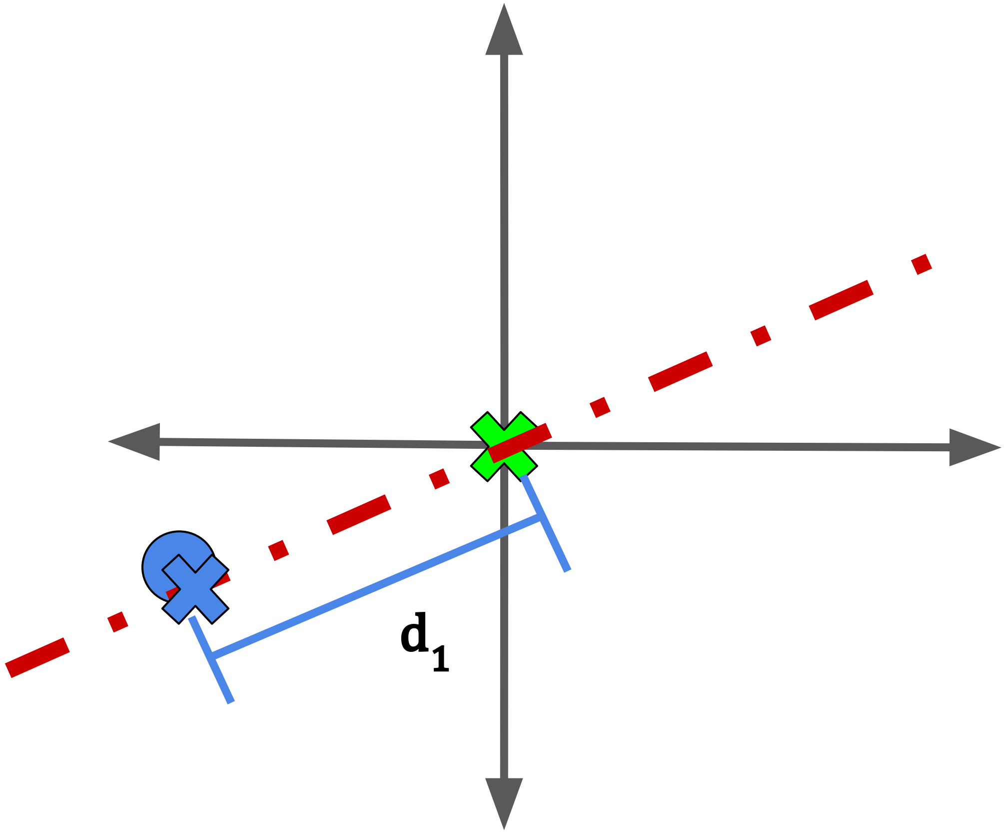

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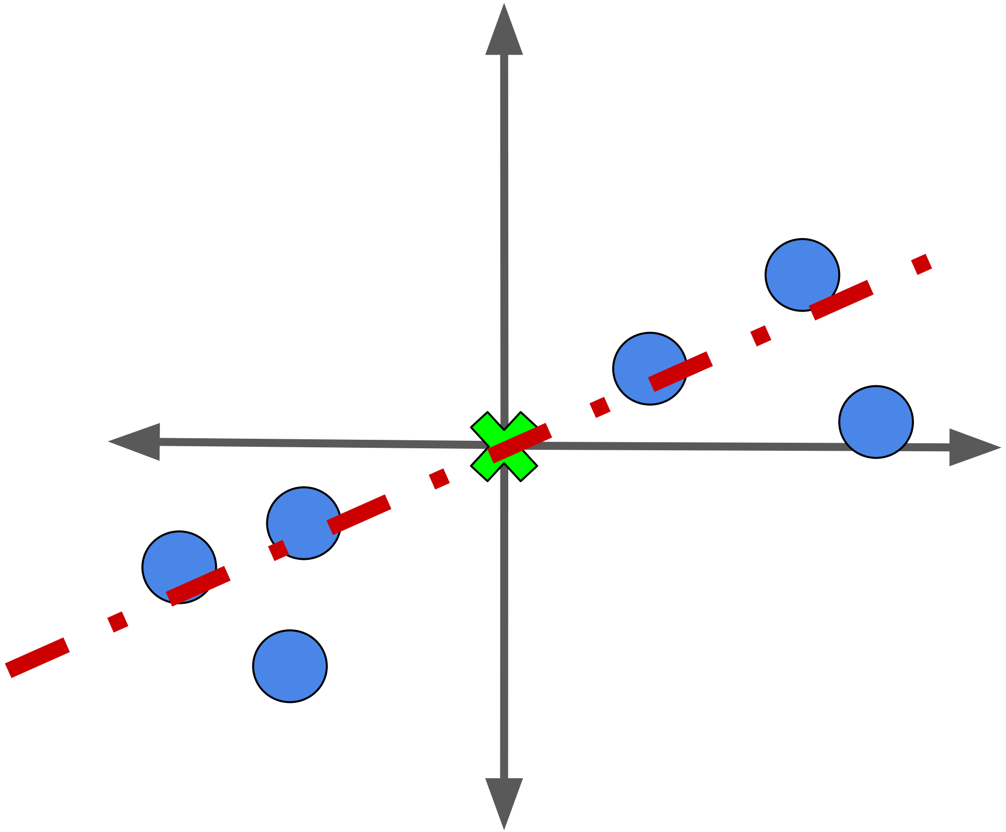
Now let us try something different: fitting a line to our data that goes through the origin.

If we do this for each data point, we can find the sum of squared distances from each point to the origin:

$$d_1^2+d_2^2+\ldots+d_n^2$$



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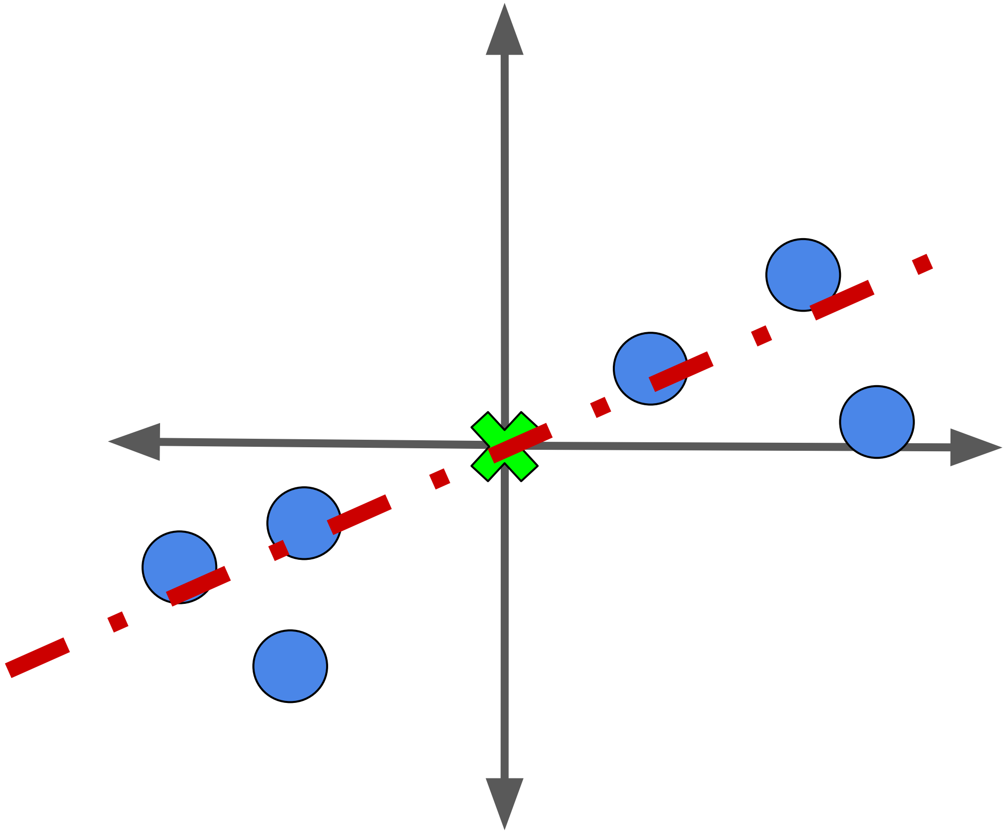
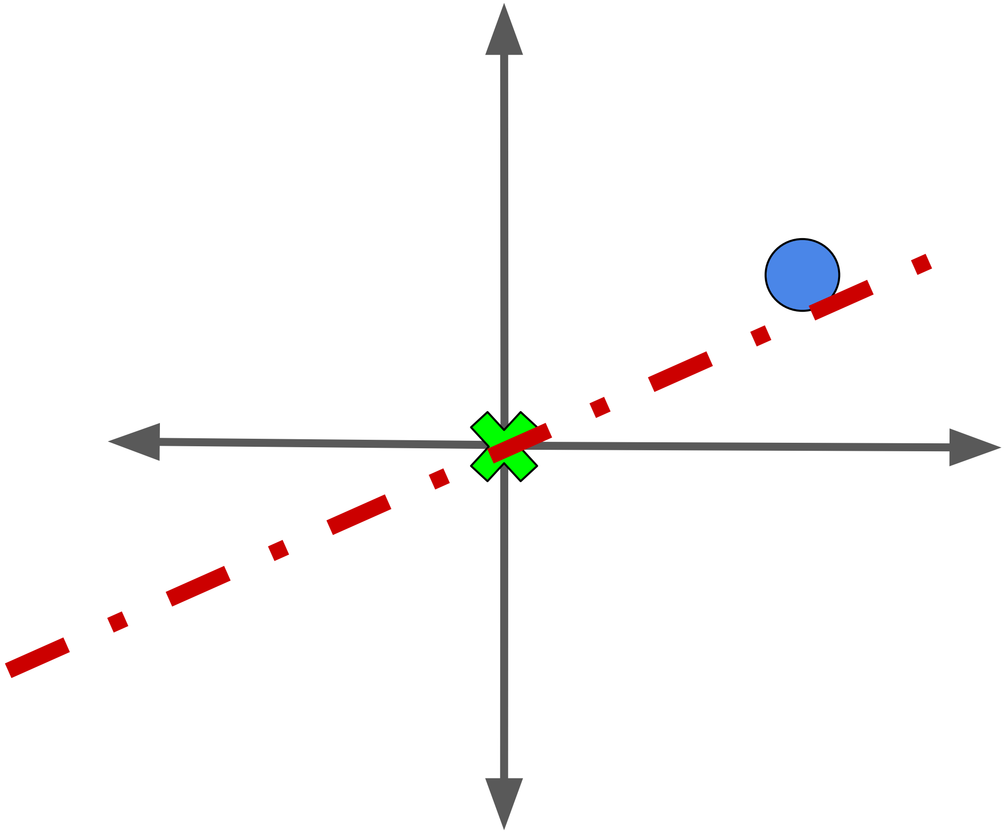
The line that maximizes this is **principal component #1**.

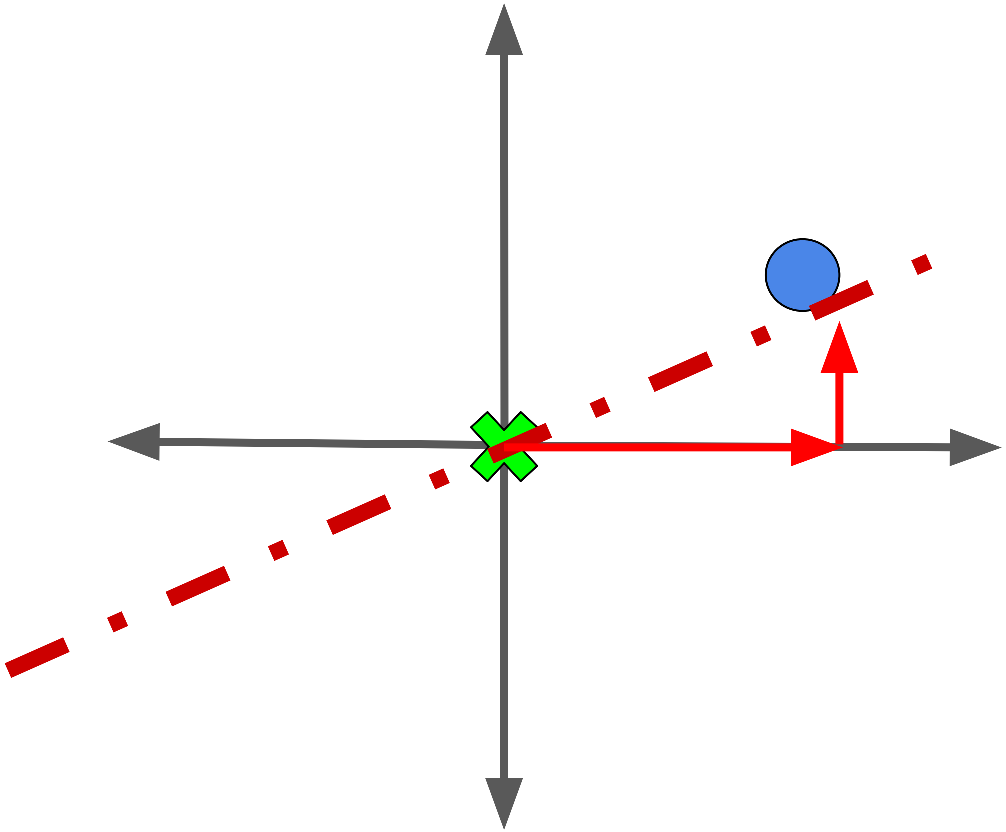


Let us again focus on just one data point.



Let us again focus on just one data point.

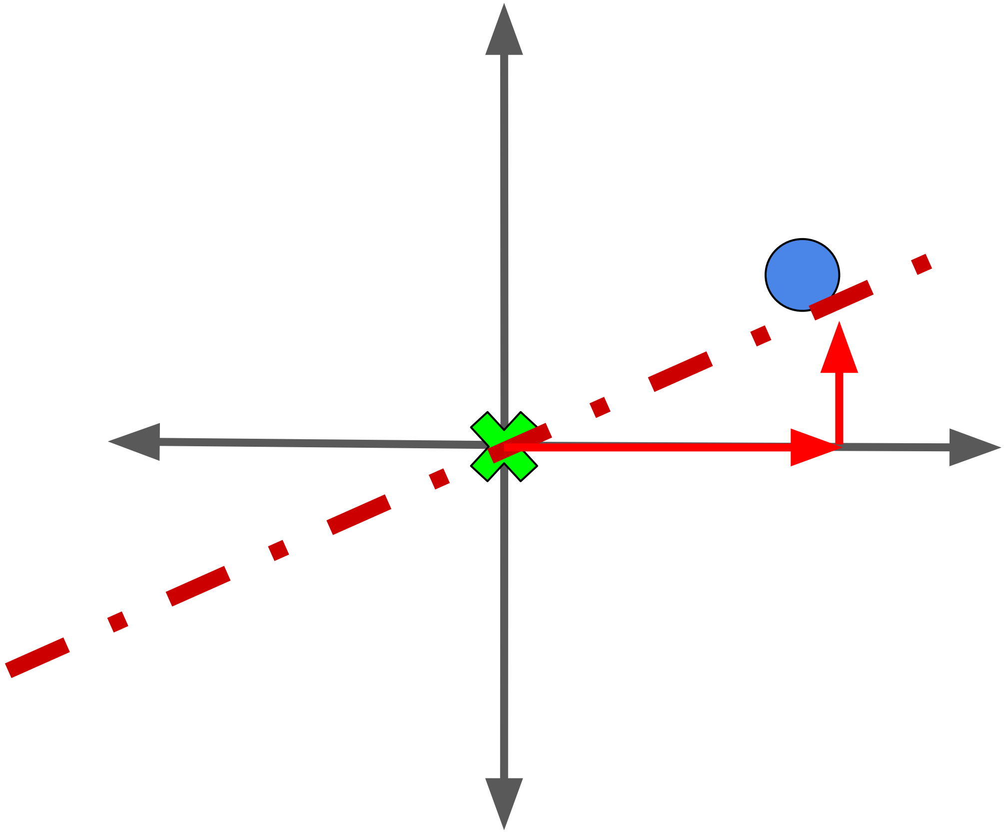
We can make a linear combination of the data point’s position on the PC by using orthogonal vectors.



Let us again focus on just one data point.

We can make a linear combination of the data point's position on the PC by using orthogonal vectors.

The length of these vectors relative to each other tells us the exact mixture of the directions needed to recover the point.



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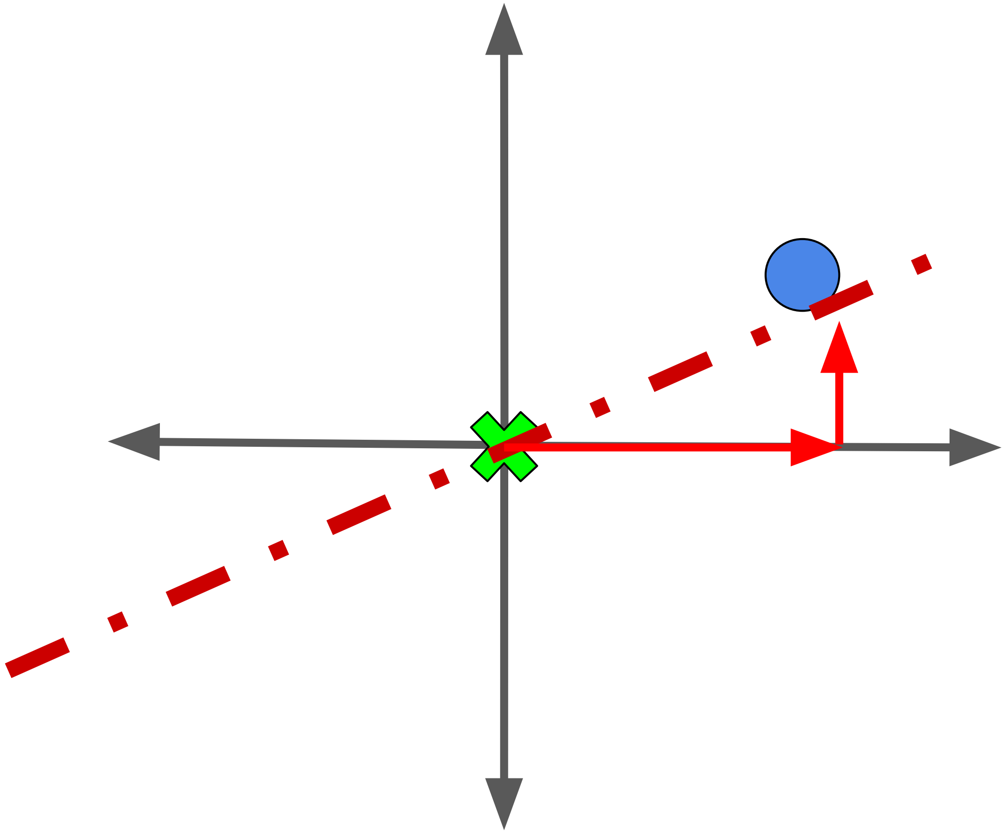
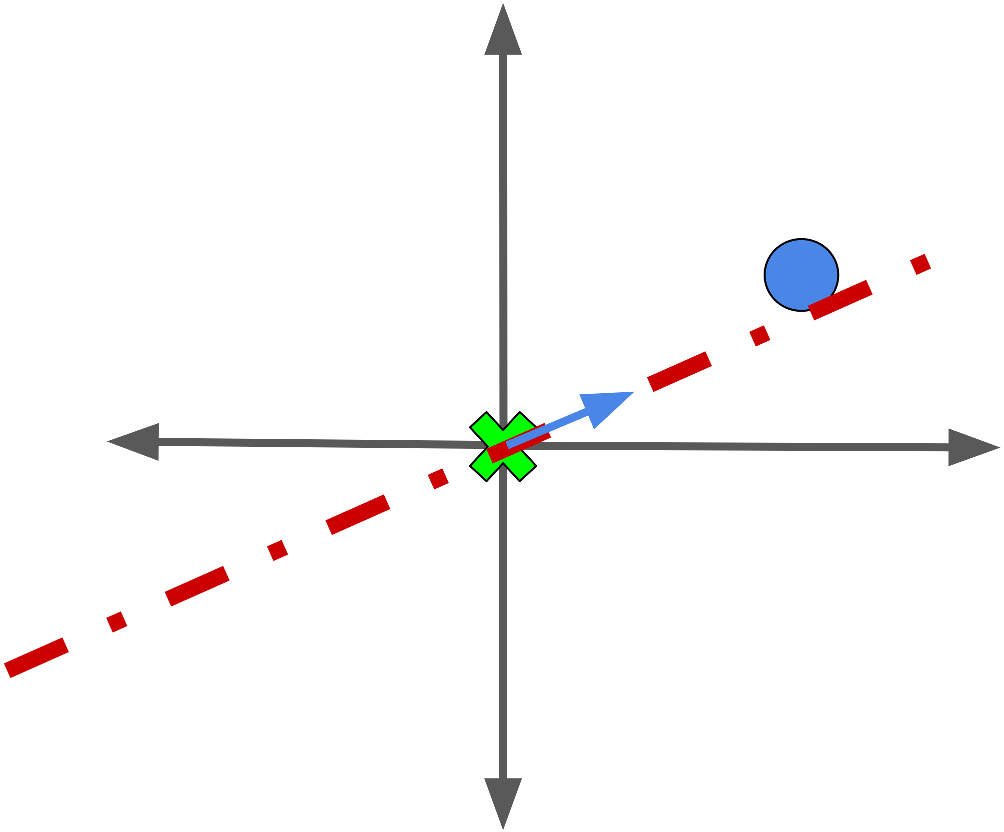
Sound familiar?



Now let's put it all together!

The unit vector going along PC #1 is its singular vector/eigenvector (U matrix).



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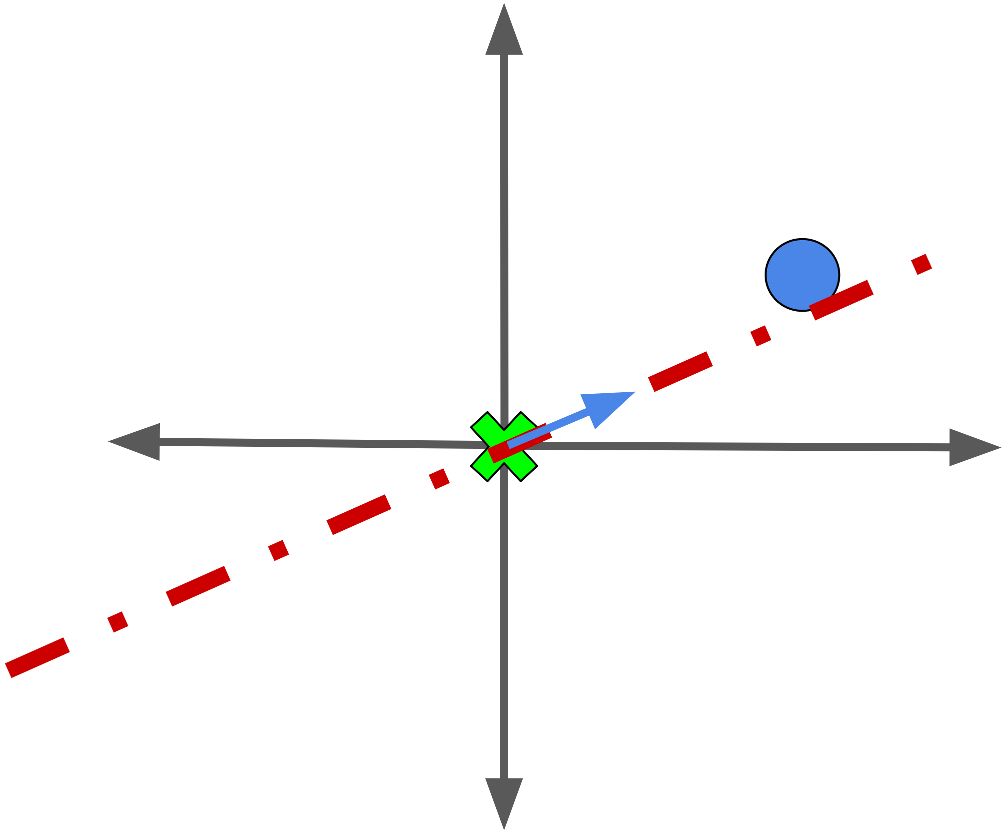
The unit vector going along PC #1 is its singular vector/eigenvector (U matrix).

The square root of the sum

$$\mathbf{d_1^2+d_2^2+\ldots+d_n^2}$$

is the singular value for PC #1 (Σ matrix).



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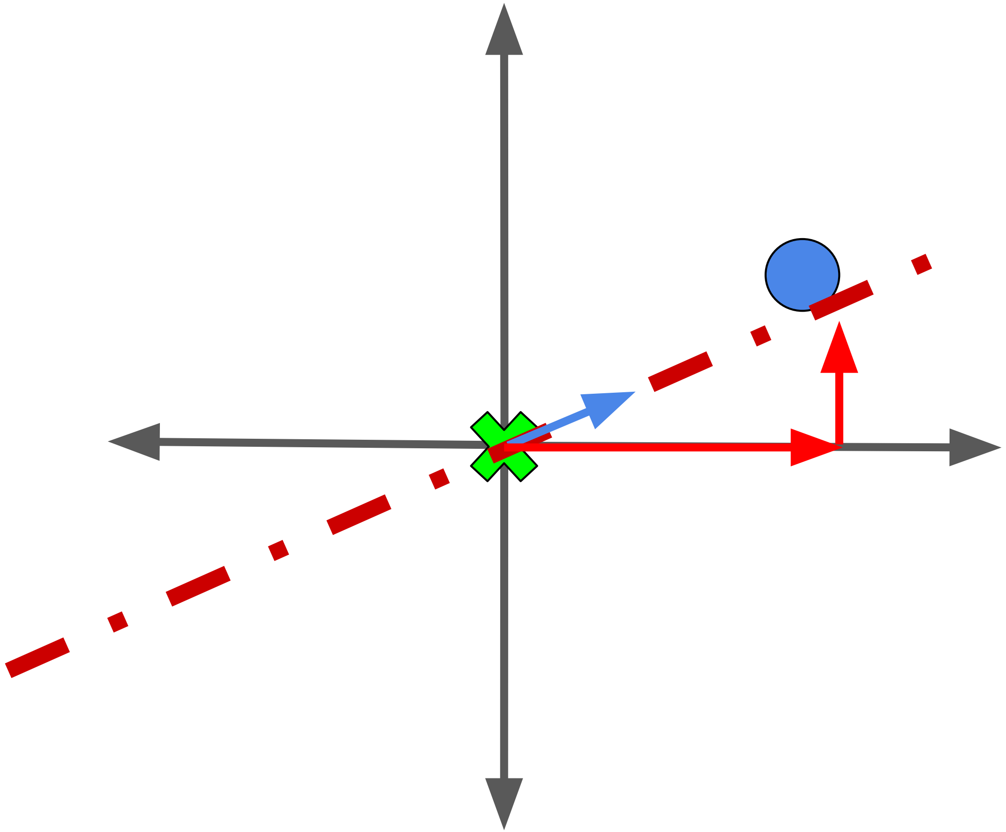
The proportions of the two features are the "loading scores" (think of V matrix).



Connecting PCA & SVD with Math

(on whiteboard)

Final Notes:

- There is a Jupyter notebook in the `stat-214-gsi` repo which gives an example of performing PCA in `python`.

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- There is a Jupyter notebook in the `stat-214-gsi` repo which gives an example of performing PCA in `python`.
- Help me decide what to cover next week: